1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

TLS-rich breast tumors will exhibit improved therapy_response to adjuvant immunotherapy after neoadjuvant chemotherapy only when tumor-intrinsic programs are favorable. Specifically, high expression of MDK and/or KRT17 within TLS-rich tumors remodel TLS into dysfunctional, myeloid-dominated niches that suppress cytotoxic T cell activity, nullifying the expected benefit of TLS. In contrast, TLS-rich tumors with low MDK and/or low KRT17 will maintain supportive TLS architecture (mature germinal center–like structures, CD8+ T cells, and pro-immunogenic myeloid balance) and show enhanced therapy_response to adjuvant immunotherapy. Tumor-derived EVs carrying MDK-related cargo will contribute to TLS remodeling toward immunosuppression, whereas EVs from MDK/KRT17-low tumors will preserve TLS functionality. Systemic host state, captured by a composite index including LMR (lymphocyte-to-monocyte ratio) and HRR (a defined host inflammatory/immune-resiliency ratio), will modulate TLS effectiveness: favorable host state amplifies TLS-associated benefit, while an unfavorable state dampens it. Metabolic/hormonal factors (HDL-C, IGF-1) will further influence TLS–tumor interactions by shaping myeloid and T cell function and should be considered as secondary modifiers. Collectively, the presence of TLS will predict therapy_response only within a defined molecular and systemic context; TLS+/MDK-low/KRT17-low patients will have the highest likelihood of benefiting from adjuvant immunotherapy, whereas TLS+/MDK-high and/or TLS+/KRT17-high cases will be resistant, unless mitigated by favorable host/systemic factors.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MDK, KRT17
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MDK: MDK/KRT17/EV path (showing residual tumor programs that encode aggressive, immunomodulatory biology).

KRT17: KRT17/EV path (showing residual tumor programs that encode aggressive, immunomodulatory biology).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	EV_mRNA_cargo	Supporting modifier from knowledge graph

Secondary	LMR	Supporting modifier from knowledge graph
Secondary	HRR	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	IGF1	Supporting modifier from knowledge graph
Secondary	ALP	Supporting modifier from knowledge graph
Secondary	TG	Supporting modifier from knowledge graph
Secondary	SHBG	Supporting modifier from knowledge graph
Secondary	AI-derived_spatial_signatures	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: is linked to basal-like features and altered tumor-immune interactions. Both can influence the TLS milieu and immune cell composition.

Prediction 2: further modulate immune cell activity and TLS remodeling. In breast cancer, leveraging TCGA-BRCA as a discovery dataset allows integration of molecular (MDK, KRT17, EV signatures), histologic TLS features, and treatment outcomes to define the context in which TLS predicts therapy_response.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EV_mRNA_cargo: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LMR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.